

Qatar University
College of Arts and Sciences
Department of Mathematics and Statistics

STAT 481 — Practice Sheet 1

Data, Summary, Display, Distance

Sun, Sep 6, 2026 • in class, 50 minutes • Lectures 1–5

Instructor: Prof. Dhafer Malouche • not graded

How to read a solution — read this first

A printed solution is a record of somebody else's thinking; reading it is not producing it. Work each part *first*; stuck for ninety seconds, read only the solution's *first line*, then carry on and ask: “which line of mine would have broken if the numbers had been different?” The [tag] on each part is your revision index: the lecture, and the kind of task.

The data. worldhealth2022, the file of Lectures 2–5, loaded exactly as the lectures load it. **You have seen it before** — deliberately: this sheet revises the code you ran, on the numbers you ran it on. Thursday's quiz will not.

Problem 1 — From the file to the two matrices [L1 → L2 → L3] *load, inspect, order, $\bar{x}$, S, R*

(a) [L1 · R-OUT] The session opens the file the way Lecture 3 did:

```
[1] 175 11
'data.frame': 175 obs. of 11 variables:
 $ country      : chr  "Afghanistan" ...
 $ region       : chr  "Middle East, North Africa, Afghanistan & Pakistan" ...
 $ income       : chr  "Low income" ...
 $ life_exp     : num  65.6 ...
 $ infant_mort  : num  51.6 8.3 ...
 $ fertility    : num  4.93 ...
 $ pop65       : num  2.35 ...
 $ urban_pct    : num  25.4 ...
 $ gdp_pc       : num  357 ...
 $ health_exp_pct: num  23.1 ...
 $ internet_pct : num  15.9 ...
```

State n and p . Say what one row of $\mathbf{X}$ stands for. Name the columns that are *not* variables, and say why the lectures always write `wh[, 4:11]` rather than `wh`.

(b) [L1 · R-PREDICT] We put the country names on the rows, as Lecture 5 does, and take one indicator for the six Gulf countries:

```
W <- wh[, 4:11] # the eight indicators; columns 1-3 are labels
rownames(W) <- wh$country # country NAMES on the rows, as in Lecture 5
gcc <- c("Bahrain", "Kuwait", "Oman", "Qatar", "Saudi Arabia",
        "United Arab Emirates")
v <- W[gcc, "infant_mort"]; names(v) <- gcc
v
```

Bahrain	Kuwait	Oman
7.0	7.4	8.5
Qatar	Saudi Arabia	United Arab Emirates
5.0	5.0	4.2

Before running anything, write down what each of `sort(v)`, `order(v)`, `rank(v)` and `rank(-v)` returns — values, positions or standings; which end carries rank 1 in each of the two `rank` calls; and how the two countries that share a value will print. Then check against the block below.

United Arab Emirates	Qatar	Saudi Arabia
4.2	5.0	5.0
Bahrain	Kuwait	Oman
7.0	7.4	8.5
[1] 6 4 5 1 2 3		
Bahrain	Kuwait	Oman
4.0	5.0	6.0
Qatar	Saudi Arabia	United Arab Emirates
2.5	2.5	1.0
Bahrain	Kuwait	Oman
3.0	2.0	1.0
Qatar	Saudi Arabia	United Arab Emirates
4.5	4.5	6.0

(c) **[L2 · by hand]** Three countries on `fertility` and `urban_pct`, rounded to whole numbers so that the arithmetic is exact — `round(W[rows, cols], 0)` with the three names and the two indicators prints

	fertility	urban_pct
Oman	2	80
United Arab Emirates	1	86
Jordan	3	92

Take this 3×2 table as $\mathbf{X}$. Compute $\bar{\mathbf{x}}$, $\mathbf{S}$, r and $|\mathbf{S}|$ exactly, keeping every quantity as a fraction. Say in one sentence what the divisor $n - 1$ is doing. *Keep your $\mathbf{S}$: Problem 2 uses it.*

(d) **[L2 · R-OUT]** As Lecture 2 printed them: the diagonal of $\mathbf{S}$, then $\mathbf{R}$ at `options(width = 100)`, all 175 countries.

life_exp	67.3	infant_mort	517.7	fertility	1.6	pop65	51.0	urban_pct	487.4	gdp_pc	801635374.1
health_exp_pct	10.7	internet_pct	648.6								
life_exp	1.00	infant_mort	-0.92	fertility	0.66	pop65	0.55	urban_pct	0.56	health_exp_pct	0.17
infant_mort	-0.92	1.00	0.78	-0.56	-0.46	-0.39	-0.15	-0.74			
fertility	-0.81	0.78	1.00	-0.67	-0.50	-0.42	-0.21	-0.84			
pop65	0.66	-0.56	-0.67	1.00	0.43	0.59	0.36	0.61			
urban_pct	0.55	-0.46	-0.50	0.43	1.00	0.52	0.23	0.72			
gdp_pc	0.56	-0.39	-0.42	0.59	0.52	1.00	0.16	0.52			
health_exp_pct	0.17	-0.15	-0.21	0.36	0.23	0.16	1.00	0.20			
internet_pct	0.77	-0.74	-0.84	0.61	0.72	0.52	0.20	1.00			

- Off the diagonal, name the strongest association in $\mathbf{R}$ and the weakest, with values, and say in words what the sign of the strongest means.
- The eight variances add up to $\text{tr}(\mathbf{S})$. Why does `gdp_pc` own 100% of it to two decimals, and what happens to that share if income is recorded in *thousands* of dollars (`gdp_pc / 1000`)? Direction and reason; a number is a bonus.

(e) **[L3 · R-WRITE]** Write the R that prints the mean of *each* of the eight indicators within *each* income group, one table, groups in economic order (Lecture 3's `econ` vector). Then name the one indicator whose four group means do not run in one direction as income rises.

(f) **[L2 · concept]** An economist wants to know how many dollars of income go with one extra year of life expectancy; a journalist wants to know which two of the eight indicators are most tightly linked. Which matrix does each one get, and why? Then: two pairs have $r = 0.6$ and $r = -0.6$ — what does $|\mathbf{R}| = 1 - r^2$ say about them, and what does it miss?

Problem 2 — Which countries are unusual, and by which yardstick? **[L4 → L5]** *displays, a 2×2 , the workflow, two rankings, code that lies*

(a) **[L4 · choosing a display]** An analyst at the Ministry of Public Health asks three questions in turn. For each, name the display from Lecture 4 you would draw first, and say what that display *cannot* show. Do not draw anything.

- (i) “Does life expectancy keep rising with income, or does it level off?”
- (ii) “Of the 52 indicators in the wide file, which ones are telling me the same thing?”
- (iii) “Across all eight indicators at once, what does Qatar *look like* next to its 17 MENA neighbours?”

(b) **[L5 · by hand]** Use the $\mathbf{S}$ of Problem 1(c), $\mathbf{S} = \begin{pmatrix} 1 & 3 \\ 3 & 36 \end{pmatrix}$, and a fourth country whose deviation from the mean is $\mathbf{x} - \bar{\mathbf{x}} = (1, -6)^\top$: one more child per woman, six points less urban. Take $\mathbf{S}$ as given.

- (i) Compute $\mathbf{S}^{-1}$ in exact fractions, and check it: multiply row 1 of $\mathbf{S}$ entry-by-entry against column 1 of $\mathbf{S}^{-1}$ and add.
- (ii) Compute $d^2(\mathbf{x}, \bar{\mathbf{x}})$ exactly.
- (iii) Compute the squared Euclidean distance of the same deviation, and say in one sentence why the two numbers differ.

(c) **[L5 · R-PREDICT]** We now hunt unusual countries on all eight indicators, first dropping the one row Lecture 3 taught you to drop:

```
ok <- wh$life_exp > 40                                # Lecture 3, step 5: the broken row
X <- as.matrix(wh[ok, 4:11]); rownames(X) <- wh$country[ok]
dim(X)
```

Before running anything: what is the *first* line of the Lecture 5 workflow, and will a raw straight-line `dist(X)` be driven by one column — which?

`dim(X)` and then `round(apply(X, 2, sd), 2)` print

```
[1] 174      8
      life_exp      infant_mort      fertility      pop65      urban_pct
      7.13         17.83         1.23         7.14         22.10
      gdp_pc health_exp_pct      internet_pct
      28361.75      3.27         25.18
```

(d) [L5 · R-INTERP] Both rankings measure every country from the **centre** $\bar{x}$ of the eight-indicator cloud — one by statistical distance, one by the raw straight line — and print the six farthest:

```
d2 <- mahalanobis(X, colMeans(X), cov(X)) # from the centre, all 174
round(sort(d2, decreasing = TRUE)[1:6], 2)
eu <- sqrt(rowSums(sweep(X, 2, colMeans(X))^2)) # straight line, Lecture 5
round(sort(eu, decreasing = TRUE)[1:6], 1)
c(by_d2 = rank(-d2)[["Qatar"]], straight_line = rank(-eu)[["Qatar"]])
```

Monaco	Naoero	Afghanistan	Somalia, Fed. Rep.
75.52	47.67	46.15	29.55
Lesotho	Qatar		
25.60	21.15		
Monaco	Luxembourg	Norway	Ireland
207497.8	105165.4	94567.9	86636.4
by_d2	straight_line	Switzerland	Singapore
6	7	79254.8	72673.4

Where does Qatar stand on each list? Then, in **three sentences** for the Ministry analyst: why do the two lists barely overlap, and which one to trust?

(e) [L5 · concept] State exactly when the Euclidean and the Mahalanobis distances return the *same* distance matrix, and justify it in two lines.

(f) [L5 · R-FIX] The lines below were written to answer: *which three countries are farthest from the centre of the eight-indicator cloud, in statistical distance?* They do not do that. (`t()` is R's transpose: it tips a table on its side, swapping its rows and its columns.)

```
1 wh <- read.csv("data/worldhealth2022.csv", row.names = 1)
2 X <- as.matrix(wh[wh$life_exp > 40, 4:11])
3 S <- cov(t(X))
4 d2 <- scale(mahalanobis(X, colMeans(X), solve(S)))
5 sort(d2, decreasing = TRUE)[1:3]
```

Run as printed, R stops with

```
Error in solve.default(S) :
  system is computationally singular: reciprocal condition number = 2.77647e-24
```

Name, by line number, and correct every line that is wrong, and say what each wrong line actually computes. Then: the error message points at one line — is that the line that is wrong?

Problem 3 — Fill in the blanks [L2 → L5 · quiz rehearsal] *laptops closed*

If the clock reads 33 and this problem has not started, skip to it. The transcript below is one R session on `worldhealth2022`, run exactly as Lectures 2–5 ran it; every line is correct as printed, and pieces of it are hidden behind numbered yellow boxes. Write what each box hides — a function name, an argument, a number or a name — in the answer table on the next page. Everything you need is on this page: a hidden number is always one that another line of the transcript, with what you know from Lectures 1–5, lets you recover. One point per box.

```
> wh <- read.csv("data/worldhealth2022.csv", (1))
> dim(wh)
[1] 175 11
> ok <- wh$life_exp > 40 # Lecture 3, step 5: drop the broken row
> sum(ok)
[1] (2)
> X <- (3)(wh[ok, 4:11])
> rownames(X) <- wh$country[ok]
> dim(X)
[1] 174 (4)
> round((5)(X), 2)
      life_exp infant_mort fertility pop65 urban_pct
      72.83    18.91      2.44    9.78    61.73
      gdp_pc health_exp_pct internet_pct
      18554.28      6.80      69.89
> round(cov(X)[1:4, 1:4], 2)
      life_exp infant_mort fertility pop65
life_exp    50.90    -113.70    -7.30  36.53
infant_mort (6)      317.77    18.60 -83.90
fertility    -7.30    18.60    1.52 -5.92
pop65        36.53    -83.90 (7)    50.99
> round(cor(X)[1:4, 1:4], 2)
      life_exp infant_mort fertility pop65
life_exp    1.00    -0.89    -0.83  0.72
infant_mort -0.89    1.00    0.85 -0.66
fertility    (8)      0.85    (9) (10)
pop65        (11)    -0.66 (12) 1.00
> round(apply(X, (13), sd), 2)
      life_exp infant_mort fertility pop65 urban_pct
      (14)      (15)      1.23    7.14    22.10
      gdp_pc health_exp_pct internet_pct
      28361.75      3.27      25.18
> Z <- (16)(X)
> round(cov(Z)[1:3, 1:3], 2)
      life_exp infant_mort fertility
life_exp    1.00    (17)    (18)
infant_mort -0.89    (19)    0.85
fertility    -0.83    (20)    1.00
> (21)(Z[c("Qatar", "Kuwait", "Bahrain", "United Arab Emirates"), ])
      Qatar Kuwait Bahrain
Kuwait    1.8562957
(22)      2.1483172 0.4388862
United Arab Emirates 1.7262102 0.8643204 1.1809634
> d2 <- mahalanobis(X, colMeans(X), (23))
> round(sort(d2, (24) = TRUE)[1:6], 2)
      Monaco Naoero Afghanistan Somalia, Fed. Rep.
      75.52    47.67    46.15    29.55
      Lesotho (25)
      25.60    21.15
> (26)(-d2)[["Qatar"]]
[1] 6
> XW <- X %*% solve(chol(cov(X))) # Lecture 5, the whitening line
> all.equal(as.matrix(dist(XW))["Qatar", "Kuwait"]^2,
+ (27)(X["Qatar", ], X["Kuwait", ], cov(X)))
[1] (28)
```

Answer table — Problem 3. One entry per box; write exactly what the box hides.

box	contents	box	contents
(1)		(15)	
(2)		(16)	
(3)		(17)	
(4)		(18)	
(5)		(19)	
(6)		(20)	
(7)		(21)	
(8)		(22)	
(9)		(23)	
(10)		(24)	
(11)		(25)	
(12)		(26)	
(13)		(27)	
(14)		(28)	

△! The three errors this sheet exists to prevent

1. **Reading a covariance as if it were a correlation**, or comparing variances in different units (Problem 1(d), (f)): **S** carries the units, **R** does not.
2. `mahalanobis(X, xb, solve(S))` (Problem 2(f)): no error, no warning, and not d^2 .
3. `cov(t(X))`, and **a distance on raw mixed-unit columns** (Problem 2(c), (d), (f)): both run, both answer a question nobody asked. Say the dimensions and units aloud before Enter.

Quiz 1: Thursday, Sep 10, in class, Lectures 1–5, fresh data, no R.